**CEG5302 Group Project - Group 01**

## Method

We applied the group's NSGA-II implementation to ZDT3 and vehicle crashworthiness, then used constrained NSGA-II on MW7 and RCM20. Selection combines non-dominated rank and crowding distance; simulated binary crossover and polynomial mutation generate offspring. For the constrained problems, feasible points outrank infeasible points, and infeasible points are ordered by total constraint violation.

## Task 1. ZDT3

The 30-variable benchmark has a disconnected true front [1]. With a population of 100 and max_gen=500, the final objective-space scatter covers all five separated segments (Fig. 1). Crowding distance favours sparsely sampled regions, which can help retain coverage across the disconnected front.

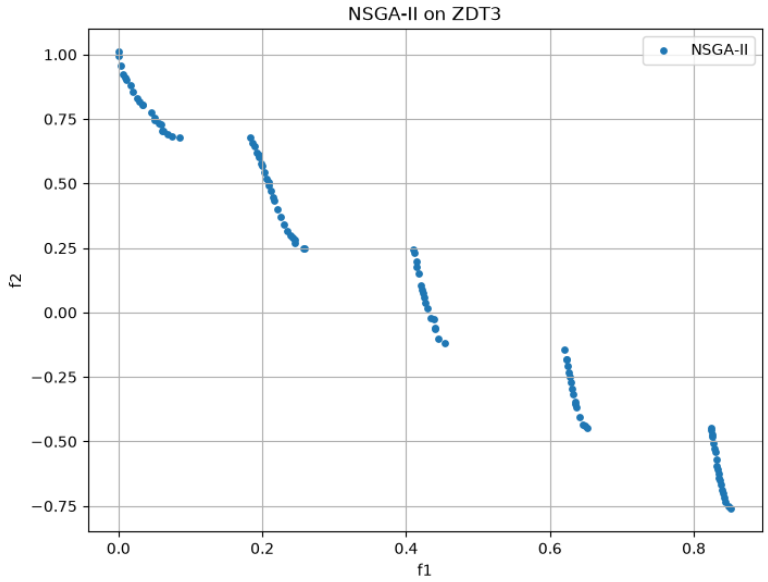



*Fig. 1. ZDT3 final population from the group's NSGA-II run.*

## Task 2. Vehicle crashworthiness

We solved the five-variable crashworthiness model [2] with a population of 100, max_gen=500 and seed 2026. NSGA-II returned 100 designs (98 distinct). Weight, acceleration and toe-board intrusion trade off: designs with lower intrusion can have greater weight or acceleration. The pairwise views show clusters and sparsely sampled gaps (Fig. 2). The nonlinear mapping from five thickness variables to three objectives and crowding distance acting only among sampled solutions can both contribute to uneven spacing. Larger populations or stronger mutation should be compared across seeds at similar evaluation budgets.

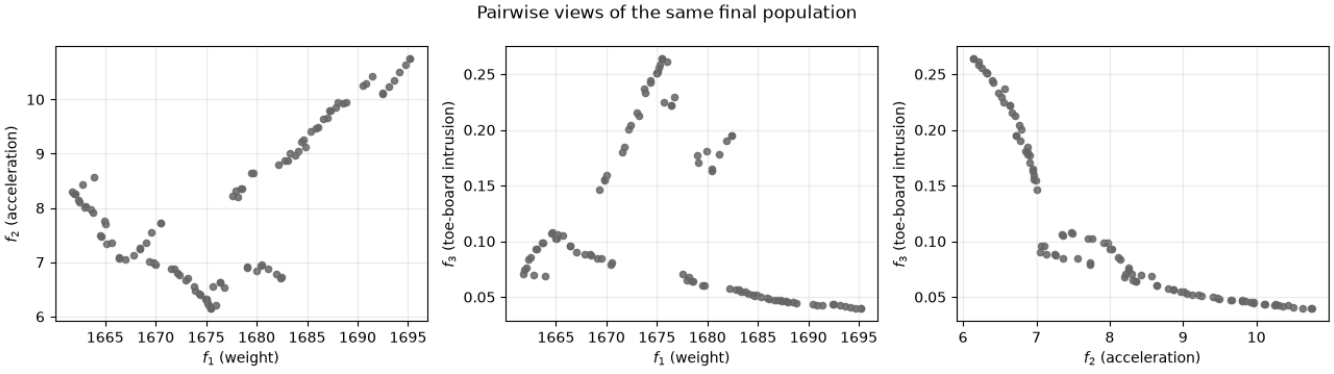



*Fig. 2. Pairwise projections of the crashworthiness final population (100 designs).*

### Task 3. Constrained NSGA-II on MW7

For MW7 [3], five seeds (0-4) used N=100, 500 evolutionary updates, crossover probability 0.8 and per-variable mutation probability 0.05. Every final population was fully feasible, yet only seeds 0, 3 and 4 covered all three visible branches. Seeds 1 and 2 stayed on the upper-left branch (Fig. 3). Once feasible solutions are plentiful, feasibility-first selection rejects infeasible candidates; near-parent variation can then keep the search in a reached region. Crowding distance distributes points along that region but cannot identify an undiscovered branch. More exploratory mutation and a larger population should be assessed by branch coverage across multiple seeds, not just the count of feasible or non-dominated solutions.

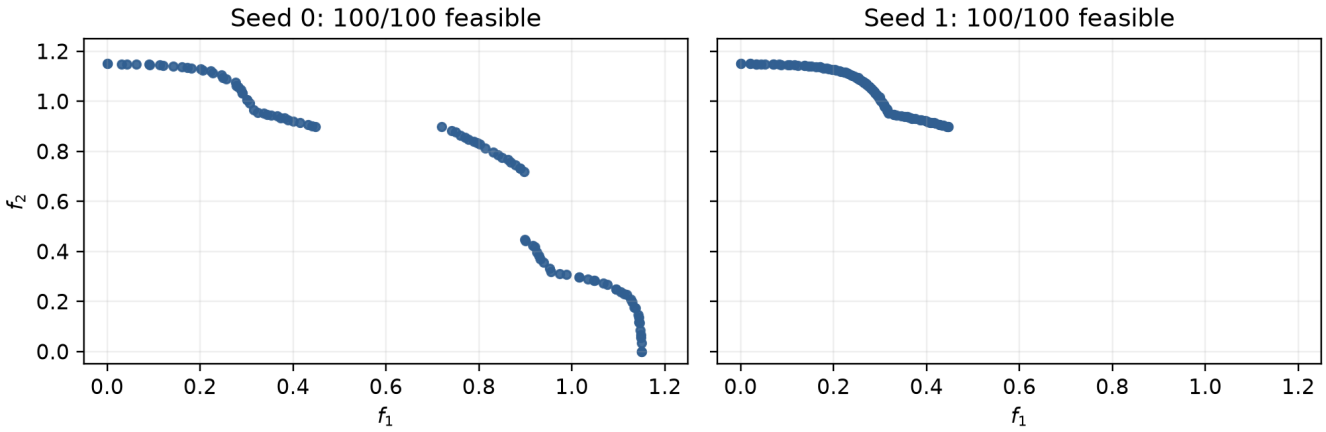



*Fig. 3. MW7 final populations at seeds 0 and 1 on identical objective scales. Both are feasible; seed 1 misses two branches.*

### Task 4. RCM20 model and experiment

RCM20 is a four-variable, two-objective bearing-design problem with seven inequality constraints [4]. We implemented the authors' MATLAB *CEC2021_func.m* branch for RCM20 [5], whose equations differ from the printed supplement: the code requires pressure P0 at most 1000, while the supplement prints the opposite inequality. Constraint violation is the sum of positive violations. One-factor sweeps varied population size N, generation limit G, crossover probability pc and mutation probability pm over five seeds. A second stage compared N-G splits near 200,000 evaluations; a third stage combined the best levels at exactly 200,000 evaluations. The setting selected by median HV with a diversity tie-break was tested on ten new seeds (100-109).

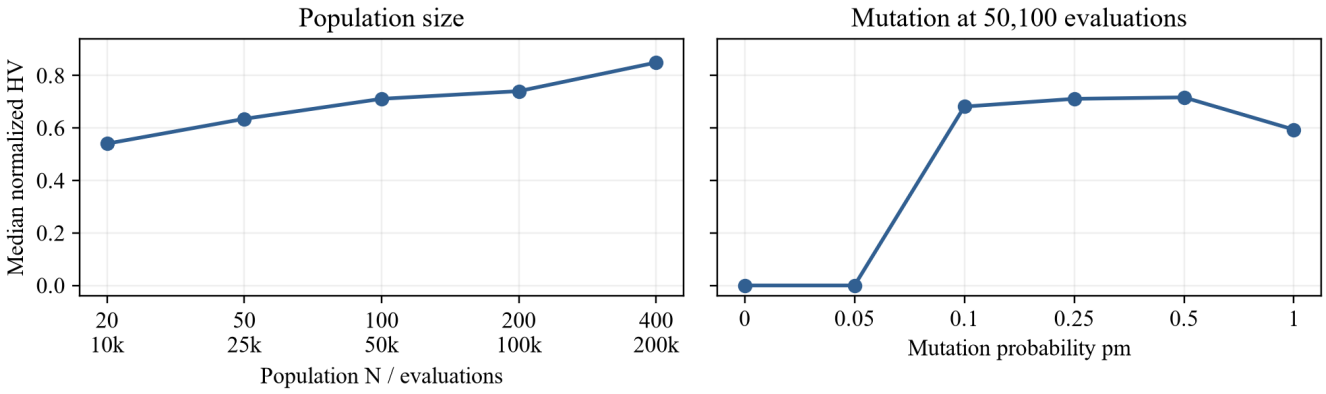



*Fig. 4. RCM20 Stage A median HV over five seeds. Increasing N also increases evaluations; the pm sweep holds evaluations fixed.*

**Task 4. RCM20 results**

Mutation had the largest observed effect: with pm=0, three of five runs found no feasible solution, while pm=0.1-0.5 yielded median normalized hypervolume (HV) of 0.68-0.72. Increasing N or G improved HV in the one-factor sweeps but also increased evaluations. Near 200,000 evaluations, different N-G splits gave median HV of 0.76-0.85. Crossover effects varied with N and pm.

| Configuration | Feasible runs | Median HV | Median distinct |
|---|---|---|---|
| Selected: N=400, G=499 | 10/10 | 0.839 | 261.5 |
| Equal-budget baseline: N=100, G=1999 | 9/10 | 0.740 | 86.0 |

*Table 1. Held-out seeds 100-109. Both settings use 200,000 objective evaluations; pc=0.9 and pm=0.25.*

The selected configuration had higher HV on seven of ten paired seeds and produced more distinct solutions, although its median largest spacing gap was also larger (0.136 versus 0.051). Pooling its ten runs and removing duplicates yielded 598 feasible, mutually non-dominated designs (Fig. 5). A known feasible reference point dominates the obtained lowest-f1 extreme, indicating room for further convergence at that end.

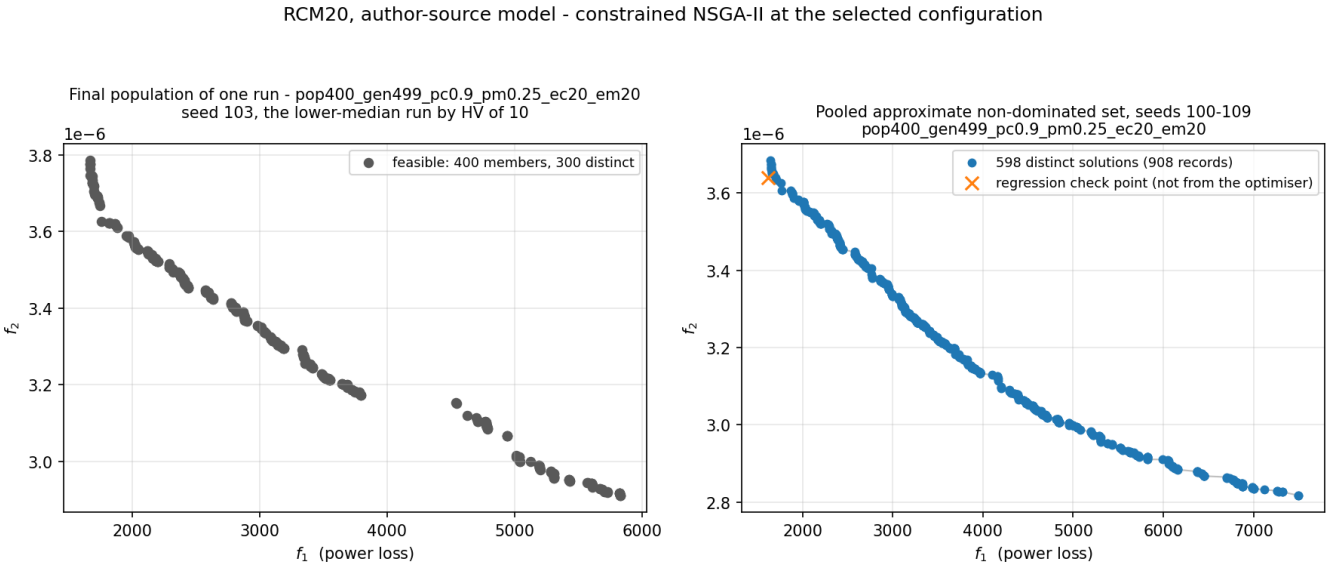



*Fig. 5. RCM20 final population of seed 103 (left) and 598 distinct pooled solutions from ten seeds (right); cross: known feasible point.*

**References**


[1] E. Zitzler, K. Deb and L. Thiele, *Evolutionary Computation* 8(2), 173-195 (2000).

[2] X. Liao et al., *Structural and Multidisciplinary Optimization* 35, 561-569 (2008).

[3] Z. Ma and Y. Wang, *IEEE Transactions on Evolutionary Computation* 23(6), 972-986 (2019).

[4] A. Kumar et al., *Swarm and Evolutionary Computation* 67, 100961 (2021).

[5] P. N. Suganthan, 2021-RW-MOP GitHub repository, *CEC2021_func.m*, github.com/P-N-Suganthan/2021-RW-MOP (accessed 23 Oct 2026).